Syllabus: Mod 1: Mathematical Foundations of Data Science
3 credits

Mod 1 Dates
Mod 1 starts August 29 and runs through December 13, 2025.

Mod 1 Faculty
Professors: Wayne Snyder, Professor Emeritus of Computer Science

Mod 1 Learning Facilitators
Apoorva Gupta, Learning Facilitator

Your assigned Learning Facilitator is your main contact during Mod 1 for course-related matters pertaining to all course assignments, grades, and concerns directly arising within the module (i.e., course materials and supplies). To communicate with your Learning Facilitator, go to the BU Virtual Campus.

Throughout Mod 1, Learning Facilitators will hold **group office hours**, and students may attend as many group office hours as they wish. Group office hours will be posted weekly. Learning Facilitators will also be monitoring Piazza, the online discussion board, for questions and are available for individual appointments. An appointment can be requested by submitting a Help Request via the following link BU Virtual Campus.

Mod 1 Student Success Specialists
Annie Imperato, Learning Success Specialist.

Your Student Success Specialist will support your progression through Mod 1 and the OMDS program in general by providing valuable resources to aid your learning journey. Your Success team functions similarly to an academic coach or advisor, and this team will help you get registered and support extracurricular programming. You may reach the Student Success team at BU Virtual Campus.

Mod 1 Overview and Learning Objectives
Module 1 focuses on the mathematical foundations of data science with a dual focus on linear algebra and probability. The linear algebra component will cover: vectors, matrices, tensors, multiplication, inverses, determinants, trace and norms (L1 L2 etc.). The probability component will cover: random variables, distributions, expectation, marginal/conditional probability, independence and correlation. At the end of this module, students will be able to explain and implement linear models. You will apply these core foundational ideas in a Final Project focused on analyzing the distribution of a dataset.

During Mod 1, you will:

1. Develop a strong foundation in probability theory, including random variables, probability distributions, and the concept of expectation.
2. Interpret probabilities, including marginal and conditional probabilities, to quantify uncertain events and data.
3. Apply the concept of independence in probability, which is essential for understanding various data science models and algorithms.
4. Distinguish between vectors, matrices, and tensors, and their application in data science.
5. Perform basic operations involving vectors and matrices, such as multiplication and norms.
6. Apply linear models to existing data and generate predictions for new data.

Mod 1 Weekly Schedule

Release Dates	**Concepts**	**Live Sessions**
Week 1: August 29	Data and Uncertainty	Tuesday September 2, 5pm ET (**Tuesday session due to Labor Day)**
Week 2: September 5	Descriptive Statistics and Sampling Data	Monday September 8, 5pm ET
Week 3: September 12	What Do Models Do? and Managing Model Errors	Monday September 15, 5pm ET
Week 4: September 19	Linear Regression and Introduction to Random Variables	Monday September 22, 5pm ET
Week 5: September 26	Boolean and Continuous Random Variables	Monday September 29, 5pm ET
Week 6: October 3	Relationships Between Random Variables	Monday October 6, 5pm ET
Week 7: October 10	The Law of Large Numbers, Central Limit Theorem, and Hypothesis Testing	Tuesday October 14, 5pm ET **(Tuesday session due to Indigenous Peoples' Day)**

Week 8: October 17	Motivations for Linear Algebra	Monday October 20, 5pm ET
Week 9: October 24	Introduction to Vectors	Monday October 27, 5pm ET
Week 10: October 31	Introduction to Matrices	Monday November 3, 5pm ET
Week 11: November 7	Systems of Linear Equations	Monday November 10, 5pm ET
Week 12: November 14	Vector Spaces	Monday November 17, 5pm ET
Week 13: November 21	Linear Dimensionality Reductions	Monday December 1, 5pm ET **(note makeup session for Thanksgiving)**
November 27 – November 30	**BU Thanksgiving Recess**	**No Live Session on Monday, November 24**
Week 14: December 5	Beyond Linear Methods	Monday December 8, 5pm ET

Mod 1 Live Sessions

Live Sessions for Mod 1 are offered every Monday during the term (unless otherwise stated above), starting in Week 1. Because Live Sessions are an integral part of the program, **you are strongly encouraged to attend all Live Sessions**.

If you are unable to attend a Live Session, please watch the recording. Professors consider Live Sessions essential learning, and content covered in the Live Session is eligible to be included on Module assessments. The teaching team requests that you keep your video/web cameras on during Live Sessions (unless otherwise noted, or you may turn off your camera due to bandwidth problems).

Community Standards for Live Sessions

Live Sessions in the OMDS program are an active learning environment and include actual, real-time interactions (teaching team to student, student to student). Adherence to the Student Codes of Conduct is expected. Additionally, the program recognizes that open discourse and authentic representation of the lived experience are valuable in the learning environment. Ideas and ideologies can be framed respectfully while we all seek first to understand as well as be understood. Because OMDS encourages students to be authentic, this approach suggests that honest mistakes are possible. When this happens, learning from each other is essential. As a result, OMDS strives not to censor language or discourse during the Live Session but to demonstrate the learning opportunity as needed.

Homework Assignments

Every week in Module 1 you will complete coding homework which you will retrieve via Github and submit through Gradescope. More detailed instructions are available on the Blackboard course page. Below are details of the homework deadlines:

Homework number	Due Date
Homework 01	September 7, 11:59pm ET
Homework 02	September 14, 11:59pm ET
Homework 03	September 21, 11:59pm ET
Homework 04	September 28, 11:59pm ET
Homework 05	October 5, 11:59pm ET
Homework 06	October 12, 11:59pm ET
Homework 07	October 19, 11:59pm ET
Homework 08	October 26, 11:59pm ET
Homework 09	November 2, 11:59pm ET
Homework 10	November 9, 11:59pm ET
Homework 11	November 16, 11:59pm ET
Final Project	December 6, 11:59pm ET

Final Project

The Mod 1 Final Project consists of picking a dataset from a list of provided choices and investigating the dataset from the lenses of conditional distributions and principal components analysis.

The key deliverables include:

- visualizations,
- principal components,
- analysis of the dataset distribution.

Grading

There are a total of 1,000 points in Mod 1. Throughout Mod 1, you will complete activities for credit in the various gradebook assessment categories listed below. Please make sure that you complete all your activities accordingly.

We have broken the various gradebook assessment categories out by academic content, key competencies and peer engagement.

Here's how each category contributes to your overall grade:

Category	**Grading Breakdown**
Academic Content	**100%**
• Assignments	80%
• Final Project	20%

Grading standards: OMDS follows Boston University grading guidelines when assigning these letter grades. The guidelines are designed to encourage rigorous standards and prohibit grade inflation across both online and in-person programs.

Policy Regarding Late Submissions

The OMDS program is committed to student success and strives to maintain the highest level of performance and integrity of the degree. All OMDS assignments (solo or team) are due at 11:59pm ET.

In the case of an emergency, students should submit a case via BU Virtual Campus to their assigned Learning Facilitator as soon as possible providing any relevant documentation The Learning Facilitator will work with the Student Success team and faculty to determine if an exception to the late submission policy can be made.

Otherwise, the following late policy applies:

- All assignments are due at 11:59pm ET for full credit with a grace period of 2 hours. The grace period only applies to the original deadline, and does not apply to subsequent partial credit deadlines.
- Assignments submitted more than 2 hours and up to 5 days late will get partial credit, but penalized 10% per day for lateness (final scores rounded to the closest full point).
- Assignments submitted more than 5 days (120 hours) late will receive 0 points.
- Specifically:
    - Within 0-2 hours of original deadline: no penalty
    - Within 2-24 hours of original deadline: 90% of grade
    - Within 24-48 hours of original deadline: 80% of grade
    - Within 48-72 hours of original deadline: 70% of grade
    - Within 72-96 hours of original deadline: 60% of grade
    - Within 96-120 hours of original deadline: 50% of grade
    - More than 120 hours late: 0% of grade

Projects and exams may be subject to a different policy. Please consult the course syllabi.

We follow BU's incomplete policy, as stated here:
https://www.bu.edu/academics/policies/incomplete-coursework/

Course Materials

OMDS and BU Libraries Materials

The Blackboard **"Mod 1 | What You Need to Know"** module will provide instructions on how to access learning materials through BU Libraries that will help you throughout Mod

You should plan to revisit the What You Need to Know folder in BB frequently throughout the module.

Technology Advice for OMDS

For the best technical experience, students are encouraged to use a laptop or desktop computer running Mac OS X or Windows. Recommended browsers include Firefox or Chrome – access this link for more information on browser compatibility and other system requirements. In the event of technical difficulties, you can try an alternate browser or submit a help desk ticket to request help troubleshooting the problem.

While course materials are accessible from mobile devices, not all tools are optimized for these devices, and we would strongly recommend using a laptop or desktop to complete assignments and other key components of your coursework.

Requesting Accommodation

In accordance with university policy, every effort will be made to accommodate unique and special needs of students with respect to speech, hearing, vision, or other disabilities. Any student who feels they may need accommodation for a documented disability should contact BU Disability & Access Services online or by phone at 617-353-3658 for review and approval of accommodation requests. It is best to start this process as early as possible.

Sexual Misconduct/Title IX Policy

Boston University is committed to fostering a safe learning environment for all members of its community and preventing sexual misconduct. All forms of sexual misconduct, including rape, acquaintance rape, sexual assault, domestic and dating violence, stalking, and sexual harassment, are violations of Boston University's policies, whether they happen on campus or off campus.

Title IX of the Education Amendments Act of 1972 is a federal civil rights law that prohibits sex- based discrimination in federally funded education programs and activities. This law makes it clear that violence and harassment based on sex and gender are civil rights offenses subject to the same kinds of accountability and the same kinds of support applied to offenses against other protected categories such as race, national origin, etc. If you or someone you know has been harassed or assaulted, you can find the appropriate resources at Sexual Misconduct Safety.

Academic Integrity

Students are expected to do their own work, not cheat, and must abide by Boston University's Academic Conduct Code (bu.edu/academics/policies/academic-conduct-code).

> *"Boston University is a community of learning grounded in shared values, rights, and responsibilities. One critical component of our learning community is academic integrity, that is a commitment, even in the face of adversity, to the fundamental values of honesty, trust, fairness, respect, responsibility, and courage in representing truthfully one's knowledge and ideas as one's own. Our community can only be maintained if all students represent honestly their own*



knowledge and ideas, believe that their academic competence is being judged fairly based on their knowledge and ideas, and have confidence that they will not be put at a disadvantage because of someone else's dishonesty. Promoting and protecting academic integrity is the responsibility of every member of the campus community because our goals of teaching, learning, and research can only be accomplished in environments in which ethical standards are upheld.

"Academic dishonesty undermines the bonds of trust and honesty between members of the community and defrauds those who depend upon our knowledge and integrity. Students, faculty, and staff of Boston University support this Academic Conduct Code for the purposes of providing an atmosphere of mutual trust and respect, fostering ethical behavior, and cultivating lifelong professional conduct."

Solo assignments must be completed with solo effort, and any and all work must have proper citation for all OMDS assignments. Team assignments may only be shared with your designated team. The OMDS program utilizes Turnitin (a plagiarism detection software) to assess submitted assignments. Any student believed to be in violation of the Academic Conduct Code for any reason will be subject to the university process as outlined at Academic Conduct Code.

Students should learn how to use AI text generators and other AI-based assistive resources (collectively, AI tools) to enhance rather than damage their developing abilities as writers, coders, communicators, and thinkers. The CDS Generative AI Assistance (GAIA) Policy provides guidance on how to use AI tools safely and responsibly while adhering to the Academic Conduct Code.

Community Code of Conduct

As members of the OMDS community, each of us has a collective and individual responsibility to conduct ourselves in accordance with Boston University's core values:

1. Collaborate with each other and the teaching team (i.e., faculty and Learning Facilitator team) to foster a culture of inclusion.
2. Treat each other with dignity, fairness, and respect.
3. Communicate using a professional tone and clarity to model the behavior of a business leader.
4. Debate respectfully in online forums.
5. Give feedback directly to others respectfully and constructively.

Violations of the community code of conduct pose a challenge to the overall student code of conduct and may be subject to administrative response and consequence. Students are valued members of the OMDS community and can be assured that OMDS faculty and staff also adhere to the same community code of conduct.

*The OMDS program reserves the right to modify the requirements of the Mod with adequate notice to students and within the scope of authority and responsibility entrusted by Boston University.